

SHOW COMPLETE SOLUTION FOR EACH PROBLEM.

1. SIMPSON' S VS ROMBERG VS GAUSSIAN QUADRATURE (/10)

Given the integral $\int_0^1 \frac{1}{\sqrt{2\pi}} e^{-x^2} dx$

- approximate using Composite Simpson's with 5 points
- approximate using Romberg's R_{23}
- approximate using Gauss-Legendre quadrature using 5 points.

Nodal abscissas and respective weights are given in the right.

$\pm \xi_i$	A_i
$n = 5$	
0.000 000	0.568 889
0.538 469	0.478 629
0.906 180	0.236 927

2. MULTIPOINT METHOD (/10)

The leapfrog method is the simplest multipoint method which can be derived from the central

approximation of the first derivative given by $y'(t) = \frac{y(t+\Delta t) - y(t-\Delta t)}{2\Delta t}$. The solution at the $(k+1)^{\text{th}}$ time step of the FODE $y' = f(t, y)$ is then solved as $y_{k+1} = y_{k-1} + 2f(t_k, y_k)\Delta t$.

Given the FODE $y' = t^2 - 4y$ with initial value $y(0) = 1$, using a step size $\Delta t = 0.1$

- approximate y at $t=0.1$ using Crank-Nicholson method
- approximate y at $t=0.2$ using the leapfrog method

3. LINEAR SODE IVP (/7)

Consider the linear SODE $y'' - y' = t$ with initial values $y(0) = 0$, $y'(0) = 1$. Write the SODE in vector form and perform one step of RK4 with step size $\Delta t = 0.1$.

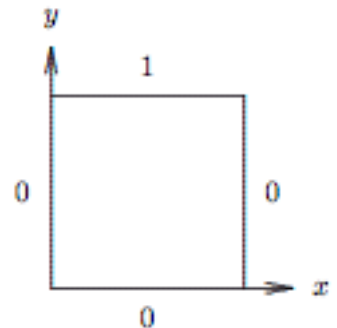
4. FDA FOR POISSON EQUATION (/8)

Starting from the Poisson equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 1$ on $0 \leq x \leq 1$ with boundary

conditions $u(x=0, y) = u(x, y=0) = u(x=1, y) = 0$ as shown in the right, derive $u(x, y=1) = 1$

the CSCS scheme (central in both spatial dimensions) for the problem and solve using

$$\Delta x = \frac{1}{3}, \Delta y = \frac{1}{2}.$$



5. CONCEPTS AND DERIVATIONS. Answer two out of the following items. (5/)

- Derive trapezoidal rule using a two-point Lagrange interpolation and the composite trapezoidal rule by dividing the interval into n panels.
- Discuss the concept of Richardson extrapolation. Derive the Richardson extrapolation formula.
- Differentiate explicit versus implicit schemes for numerical solution of differential equations and discuss how they are used for predictor-corrector schemes. Identify an example.

4. Derive the RK4 algorithm starting from $\frac{dy}{dt} = f(t, y)$.
5. Discuss the concept of adaptive step-size method and illustrate using RK4. **Outline** the algorithm for step size doubling/halving.
6. What topic/s from third exam coverage do you like the most and why?